Course Feedback

Course Outline

Finished 20%

Chapter 1.1. A Brief Introduction to AI
Chapter 1.2. AI Overview - Huawei AI Solutions
Chapter 2. AI Application Integration Requirement Analysis
Chapter 3. Huawei Cloud EI - OCR API Service
Chapter 4. Introduction to Huawei Cloud ModelArts
Chapter 5. An Introduction to AI Foundation Models and LLMs
Chapter 6. Large Model Application-AI Painting
Chapter 7. AI Application Integration Product Testing
Chapter 8. Exam Outline and Sample Questions
Section1 Exam Outline
Section2 Sample Questions

points: 30/30

True or False

Machine learning and deep learning are two independent branches of AI. They use different theories to implement AI.(1 points)

A. True

B. False✓

Answer:B

The problem abstraction in the requirement analysis prior to AI application development refers to abstracting, modeling, and converting real-world problems into problems that can be solved by AI.(1 points)

A. True✓

B. False

Answer:A

Huawei Cloud Smart Document Analysis can extract key-value pairs, recognize tables, and analyze layouts from any document image containing structured information, and return analysis results in parallel. This function is mainly built on Huawei Cloud Optical Character Recognition (OCR).(1 points)

A. True✓

B. False

Answer:A

ModelArts is a one-stop AI development platform that supports AI lifecycle development, including data processing and model training, management, and deployment. ModelArts also provides AI Gallery, which allows developers to share their models with others, promoting cooperation and knowledge sharing.(1 points)

A. True✓

B. False

Answer:A

An L2 scenario-specific model is a fine-tuned version of a domain-specific model, adapted for a specific task using a curated, labeled dataset.(1 points)

A. True✓

B. False

Answer:A

AI will completely replace humans in creative industries.(1 points)

A. True

B. False✓

Answer:B

The core innovation of the zero convolution layer is the initialized random values for parameters.(1 points)

A. True

B. False✓

Answer:B

After data cleaning is complete, further analysis and understanding of the data are needed in order to select the appropriate data for model training and iteration. This is an important step in AI application development. Understanding data can be divided into two tasks: sample distribution analysis and negative sample analysis.(1 points)

A. True✓

B. False

Answer:A

Huawei's all-scenario AI solution can be deployed in all kinds of environments, including public cloud, private clouds, various forms of edge computing, industrial IoT devices, and consumer devices.(1 points)

A. True✓

B. False

Answer:A

Single Choice

Which of the following is not a benefit of AI for creative industries?(1 points)

A. New job types

B. Low cross-domain cooperation opportunities✓

C. Faster content iteration

D. Reduced repetitive work

Answer:B

Dirty data refers to data in a dataset that does not comply with pre-defined formats or is misleading in some way. Examples of dirty data include incorrect, missing, or duplicate values, which may have a negative impact on the results of data analysis and modeling. For example, duplicate records in a subscriber database may lead to the incorrect interpretation of subscriber data. Which of the following methods can be used to handle dirty data?(1 points)

A. Model parameter tuning

B. Data cleaning✓

C. Feature extraction

D. Feature selection

Answer:B

If constraints cannot be set during the inference phase, we need to try to cover as many feature factors and use cases as possible during the training phase, so that we can identify all factors that may impact model performance while still in the training phase. For this purpose, which of the following can be used to improve the robustness and generalization performance of a model?(1 points)

A. Scenario (use case) analysis

B. Evaluation metrics

C. Data/dataset construction✓

D. Solution development

Answer:C

Which of the following is the direct function of a trainable copy in ControlNet?(1 points)

A. Replacing the original model for inference

B. Dynamically adjusting the input data distribution

C. Learning the condition control signal and generating an adjustment vector✓

D. Reducing the number of model parameters

Answer:C

Which of the following is the key component for the fusion of image and text features?(1 points)

A. Convolutional layer

B. Cross-attention layer✓

C. Pooling layer

D. Fully connected layer

Answer:B

What is the main function of timestep guidance during denoising?(1 points)

A. Adding noise

B. Reducing noise

C. Adding and reducing noise at the same time

D. Providing timestep information to optimize denoising✓

Answer:D

Which of the following capabilities is beyond the scope of today's state-of-the-art AI foundation models?(1 points)

A. Text understanding

B. Text generation

C. Protein structure prediction

D. Replacing 90% of human work✓

Answer:D

ModelArts is a one-stop development platform for AI developers. It supports major open-source AI development frameworks such as TensorFlow, PyTorch, and MindSpore, and allows developers to use self-developed algorithm frameworks. Which of the following Huawei-developed AI frameworks is supported by ModelArts?(1 points)

A. PyTorch

B. TensorFlow

C. MindSpore✓

D. PaddlePaddle

Answer:C

In the context of financial management in a company, employees often pay for business travel expenses such as airfare and hotel accommodation, and then seek reimbursement based on the invoices issued. However, filling out reimbursement forms and printing and pasting invoices can often consume employees' normal work hours. As a digitalization engineer for the company's financial processes, which Huawei Cloud AI service can be used to optimize the financial reimbursement process?(1 points)

A. Image Recognition

B. Image Search

C. Optical Character Recognition (OCR)✓

D. Speech Interaction Service (SIS)

Answer:C

Computer vision technologies can be used to fix blurry old photos. Which of the following computer vision technologies is used in this scenario?(1 points)

A. Super-resolution reconstruction✓

B. Image repair

C. Object detection

D. Image segmentation

Answer:A

Which of the following architecture can unify multi-modal data such as image, text, video, and audio?(1 points)

A. CNN

B. RNN

C. Transformer✓

D. GAN

Answer:C

John believes that in terms of knowledge representation, the concept of a cat can be represented by a "cat node" or a group of nodes that represent the attributes of cats. For example, "two eyes", "four legs", and "fluffy". Which of the following AI schools does John most likely belong to?(1 points)

A. Symbolism✓

B. Connectionism

C. Behaviorism

D. Cognitivism

Answer:A

Multiple Choices

Which of the following components are part of Huawei's full-stack AI solution?(1 points)

B. ModelArts

D. MindSpore

E. CANN

D. Kunpeng

✓

Answer:ABC

A company intends to adopt the AI development framework MindSpore for its development needs. Consequently, the company requires suitable computing devices. Which of the following options would be appropriate for purchase?(1 points)

A. Ascend NPU

B. CPU

C. GPU

D. TPU

✓

Answer:ABC

Huawei Pangu models have created a novel business model for industrial-scale AI development. They are both versatile and adaptable to a wide array of industries and scenarios. Which of the following layers are included in the hierarchical architecture of Pangu models?(1 points)

A. L0 foundation models

B. L1 industry-specific models

C. L2 scenario-specific models

D. L3 domain-specific models

✓

Answer:ABC

In formal conferences and lectures, Natural Language Processing (NLP) and speech recognition technologies can be used to automatically convert speeches into text and generate an abstract containing key meeting information. Which of the following AI technologies are used in this scenario?(1 points)

A. Speech synthesis

B. Text abstract

C. Speech recognition

D. Public opinion analysis

✓

Answer:BC

Which of the following technologies can be used for edge model optimization?(1 points)

A. Selection of a more appropriate optimizer

B. Network distillation

C. Model pruning

D. Model quantization

✓

Answer:BCD

Which of the following can be used to manage training jobs in ModelArts?(1 points)

A. Console

B. Jupyter notebook

C. Command line tool

D. DevEco Studio

✓

Answer:ABC

The self-attention mechanism is a core component of the Transformer architecture. Which of the following statements about this mechanism are true?(1 points)

A. It enables the model to capture dependencies between different positions in a sequence, regardless of distance.

B. It works by computing attention weights that determine how strongly each element in a sequence relates to every other element.

C. It can reduce model complexity.

D. It cannot capture long-range dependencies in the input.

Answer:AB

Which of the following are advantages of diffusion models over generative adversarial networks (GANs)?(1 points)

A. Higher diversity of generated samples

B. More stable training processes

C. Faster generation

D. Lower video memory usage

✓

Answer:AB

Which of the following are key factors that affect the quality of generated images?(1 points)

A. Resolution of sampling in the latent space

B. Step size of the denoising scheduler

C. UI design of the prompt text box

D. Prompt content

✓

Answer:ABD

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